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Jackson made a mistake when solving the inequality $\frac{-8x+20}{4} - 10 \geq 7$.
For questions 1- 4, use Jackson's work below.

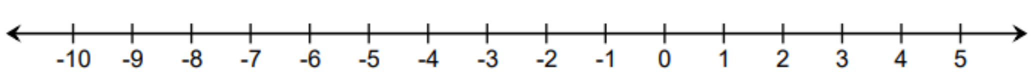
Given Inequality	$\dfrac{-8x+20}{4} - 10 \geq 7$
Step 1	$-8x + 20 - 10 \geq 28$
Step 2	$-8x + 10 \geq 28$
Step 3	$-8x \geq 18$
Step 4	$x \leq -\dfrac{18}{8}$

1. Identify the step in which Jackson made an error. Write a note to Jackson explaining why that step was incorrect.

2. Solve the inequality $\frac{-8x+20}{4} - 10 \geq 7$.

3. Write the solution in set-builder notation.

4. Graph the solution set.

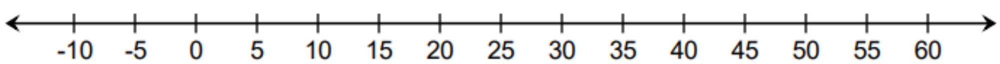


Use the inequality $0.05x + 1.13 > -0.24x + 14.18$ to complete questions 5-7.

5. Solve the inequality.

6. Write the solution in set-builder notation.

7. Graph the solution set.

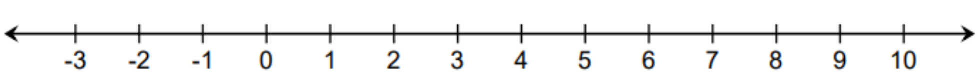


Use the inequality $-\frac{3}{2}y + \frac{2}{3} \leq -\frac{1}{3}(y + 12)$ to complete questions 8-10.

8. Solve the inequality.

9. Write the solution in set-builder notation.

10. Graph the solution set.



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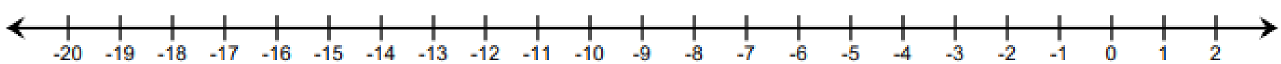
Use the following conjunctive inequality to answer questions 1-3.

$$5v + 10 \leq -4v - 17 < 9 - 2v$$

Rewrite the compound inequality as two single inequalities and solve each one.

1.	2.

3. Graph the solution set of the compound inequality.

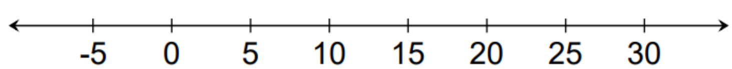


Kali just started a new sales floor job to save for college. She earns $\$15.75$ per hour plus a flat fee of $\$50$ per week. She wants to earn between $\$200$ and $\$400$ per week. The following inequality represents her earning potential:

$$200 \leq 15.75x + 50 \leq 400$$

4. Solve the inequality.

5. Graph the solution on the number line. Approximate values as necessary.



6. Complete the statement: Kali can work between _____ and _____ hours each week.

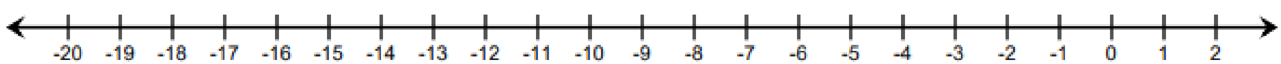
Use the disjunctive inequality shown below to answer questions 7-10:

$$-\frac{15}{4}\left(2x-\frac{2}{3}\right)<64+\frac{11}{4}x \quad or \quad \frac{35x-11}{4}\leq -134$$

Solve each component of the compound inequality.

7. $-\frac{15}{4}\left(2x-\frac{2}{3}\right)<64+\frac{11}{4}x$	8. $\frac{35x-11}{4}\leq -134$

9. Graph the solution sets of both inequalities on the number line.



10. Explain which of the following values is not a solution to the inequality.

$-20 \qquad -10 \qquad 0$

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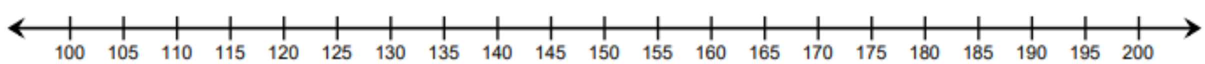
Paolo is saving for a trip to the Bahamas with his friends. He has estimated that his travel expenses will cost between $2,000 and $3,000 total. He has $250 in savings, and he earns $15.25 per hour at his job. He wants to know how many hours he must work in order to afford the trip.

1. Write an inequality to represent the situation.

2. Solve the inequality.

3. Graph the solution.



4. Check any that apply.

- ☐ Solutions must be non-negative.
- ☐ Solutions must be integers.

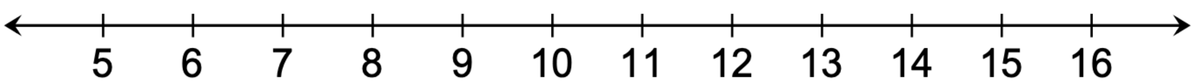
5. Explain your selection(s) from question 4.

Celine just started her own business making cake pops. She sells the cake pops in boxes, charging $\$20.00$ per box. It costs her $\$65.00$ to buy the supplies to make the cake pops. She wants to calculate how many boxes she must make to earn a profit of at least $\$225.00$.

6. Write an inequality to represent the situation.

7. Solve the inequality.

8. Graph the solution.



9. Check any that apply.

☐ Solutions must be non-negative.
☐ Solutions must be integers.

10. Explain your selection(s) from question 9.

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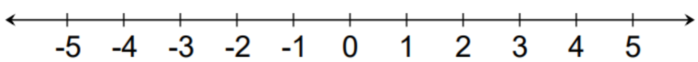
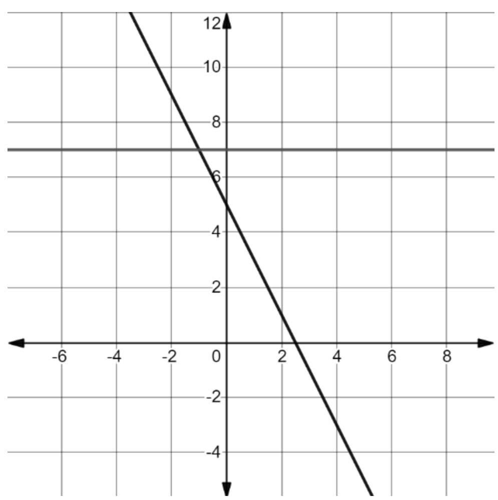
Consider the inequality $-2x + 5 \leq 7$.

1. Solve the inequality.

2. Graph the solution set.



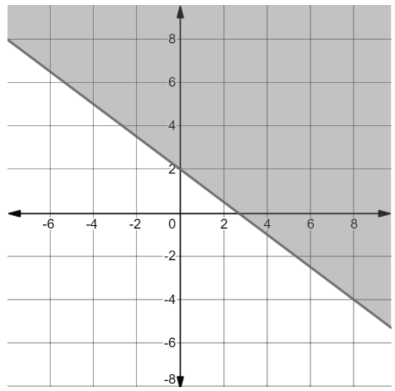
3. The graphs of $y = -2x + 5$ and $y = 7$ are shown on the coordinate plane.



Highlight the part of the graph of $y = -2x + 5$, where the y values are less than or equal to 7.

4. Describe the relationship between the solution to the inequality $-2x + 5 \leq 7$ and the values of x for which the graph of $y = -2x + 5$ is less than or equal to 7.

The graph of the two-variable inequality $y \geq -\frac{3}{4}x + 2$ is shown.



5. The slanted solid line is called the **boundary line** of the inequality. Write the equation of the boundary line in slope-intercept form.

6. How does the equation of the boundary line relate to the inequality?

7. Describe what part(s) of the graph represent the solution set of the inequality $y \geq -\frac{3}{4}x + 2$.

8. Which part of the graph of the solution set of $y \geq -\frac{3}{4}x + 2$ corresponds to each mathematical statement?

Mathematical Statement	**Graphical Representation**
$y = -3/4x + 2$	○ boundary line ○ shaded region
$y > -3/4x + 2$	○ boundary line ○ shaded region

Complete the statements.

9. If the points on the line are included in the solution set, the boundary line is
   - ○ dotted.
   - ○ solid.

10. If the points on the line are not included in the solution set, the boundary line is
   - ○ dotted.
   - ○ solid.

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The linear inequality $3x - 4y < 12$ is written in standard form.

1. Complete the statement.

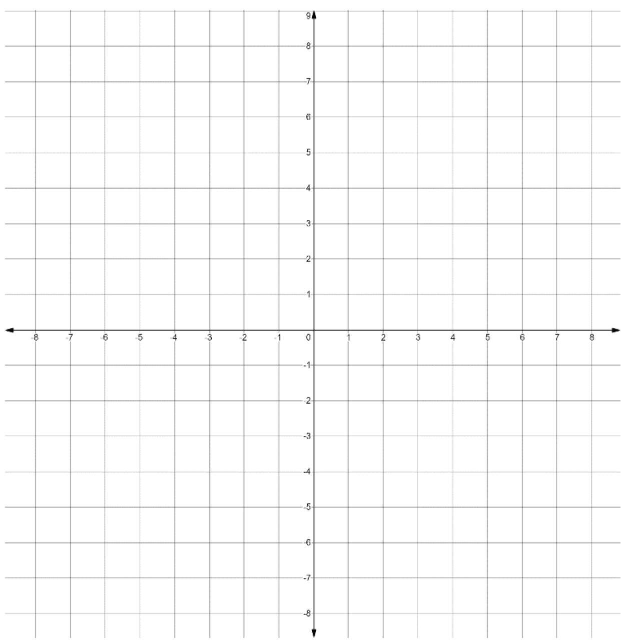
   The boundary line of this inequality is
   - ○ dotted
   - ○ solid

   because the points on the boundary line are
   - ○ included in
   - ○ excluded in

   the set.

2. Write the equation of the boundary line of this inequality in standard form.

3. Graph the boundary line on the coordinate grid.

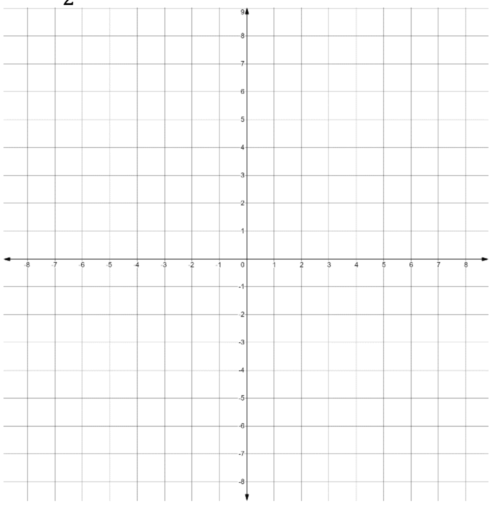


4. Explain how to determine which side of the boundary line to shade.

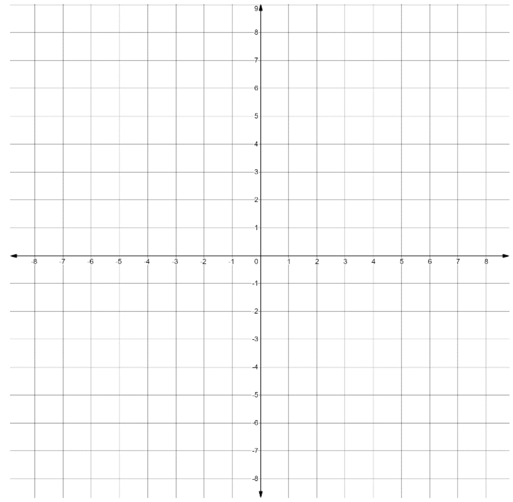
5. Shade the region on the graph that represents the solution set.

For each problem below, graph the given inequality.

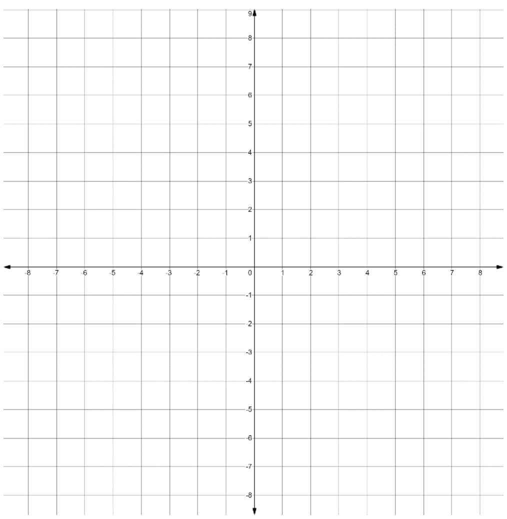
6. $y \geq \frac{1}{2}x + 2$



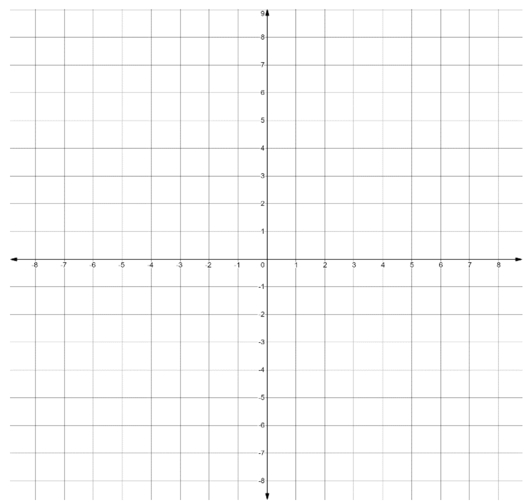
7. $y < -2x - 5$



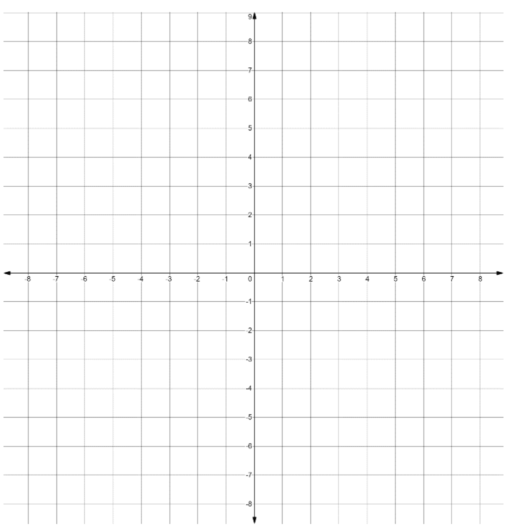
8. $x < 4$



9. $y \geq -3$



10. $y - 1 \geq -3(x + 2)$



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Molly is buying lunchmeat and cheese from the Publix deli. Lunchmeat is $\$5$ per pound and cheese is $\$4$ per pound. Molly's mother gave her $\$30$ to spend. To determine the combinations of meat and cheese that she can afford to purchase, Molly wrote the inequality $5x + 4y \leq 30$.

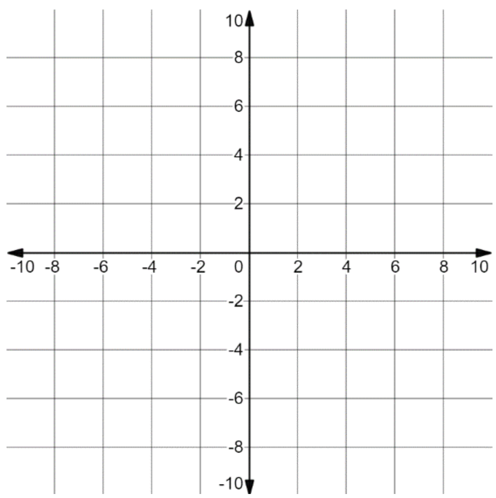
1. Define the variables Molly used based on the context.

   Let x represent ____.

   Let y represent ____.

2. Explain why the inequality sign $\leq$ makes sense in this context.

3. Graph the inequality $5x + 4y \leq 30$.



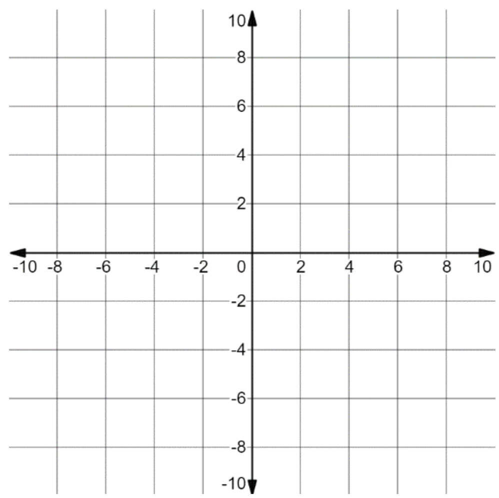
4. The ordered pair $(5, -5)$ is a solution to the inequality $5x + 4y \leq 30$. Explain why this is not a viable solution in the context of buying lunchmeat and cheese.

5. The ordered pair $(2, 5)$ is a point on the boundary line. Explain what the ordered pair represents in the situation.

Talia invested some of her savings in the stock market. She bought 50 shares of a video streaming company and 75 shares of a manufacturing company. She hopes to make a combined profit of at least $\$450$ when she sells her shares. She wrote the inequality $50x + 75y \geq 450$ to represent the situation, where x is the change in value of one share of the streaming company and y is the change in value of one share of the manufacturing company.

6. Explain what it would mean if x or y were a negative number.

7. Graph the inequality $50x + 75y \geq 450$.



8. Explain what the point $(14.75, -3.25)$ represents in Talia's situation.

9. Explain why it makes sense in Talia's context that there are no solutions for which both x and y are both negative.

10. Stock prices in the United States are rounded to the nearest hundredths place. Complete the statement:

    In order for solutions to the mathematical inequality to be viable solutions in the real-world context, the x and y values must be rounded to _____ decimal places.

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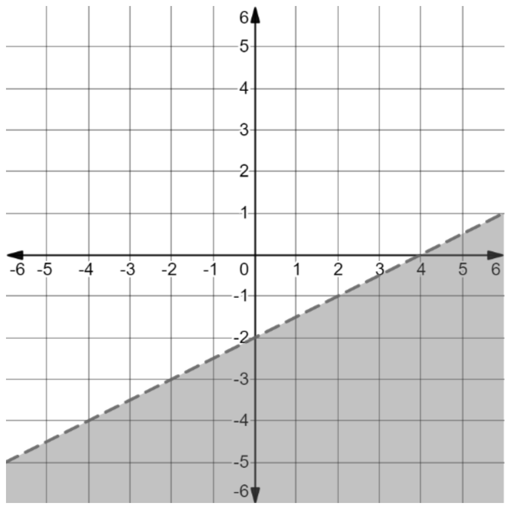
The graph of a *two* variable linear inequality is shown.



1. Write the inequality of the solution set graphed above in slope-intercept form.

2. Explain how to determine which inequality symbol to use when writing an inequality in slope-intercept form.

3. Write the inequality of the solution set graphed above in point-slope form.

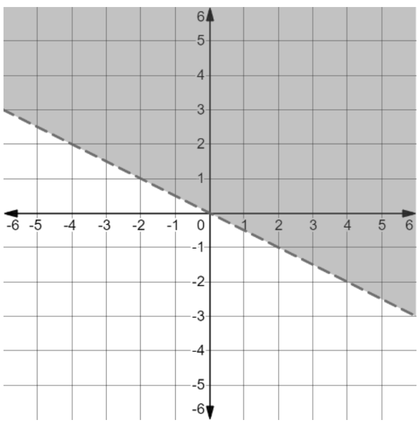
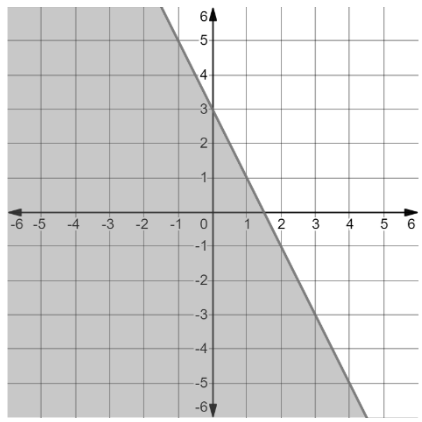
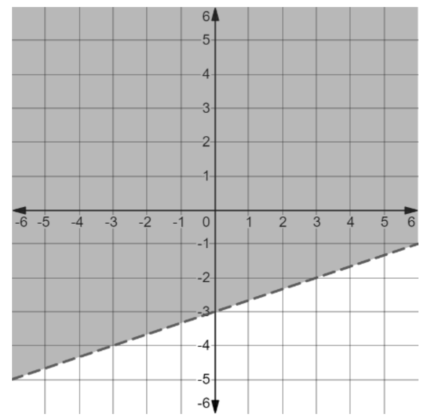
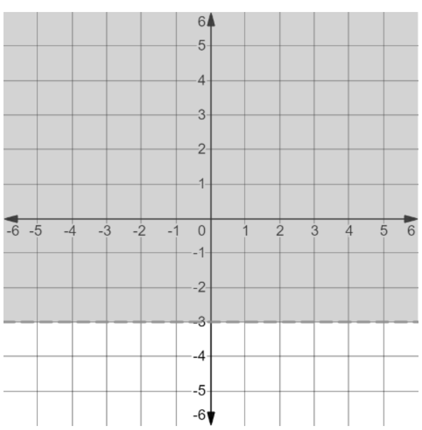
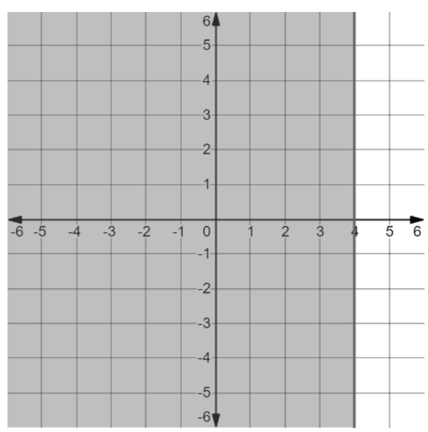
4. Explain how to determine which inequality symbol to use when writing an inequality in point-slope form.

5. Write the inequality of the solution set graphed above in standard form.

Five inequalities are given. Complete the table below to match each inequality to the graph of its solution set by writing the inequality on the line provided under each graph.

$$x \leq 4 \qquad y \leq -2x+3 \qquad -x+3y > -9$$

$$y > -3 \qquad y+2 > -\frac{1}{2}(x-4)$$

 6. Inequality: __________	 7. Inequality: __________	 8. Inequality: __________
 9. Inequality: __________	 10. Inequality: __________	

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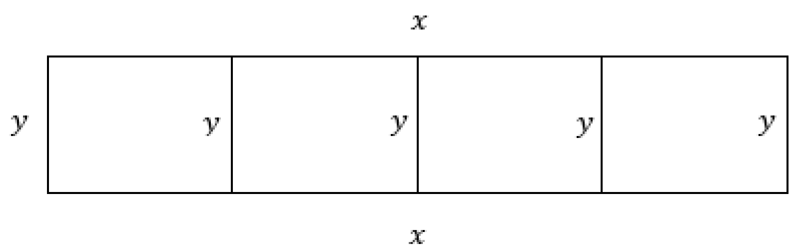
Felix wants to build a rectangular fence in his yard that will be divided into $four$ equal pens for chickens. Each pen needs to be the same size. He has 30 feet of fencing to build the fence. This scenario can be modeled as a two-variable inequality.

1. Describe what the two variables would represent in the context of Felix's fence.

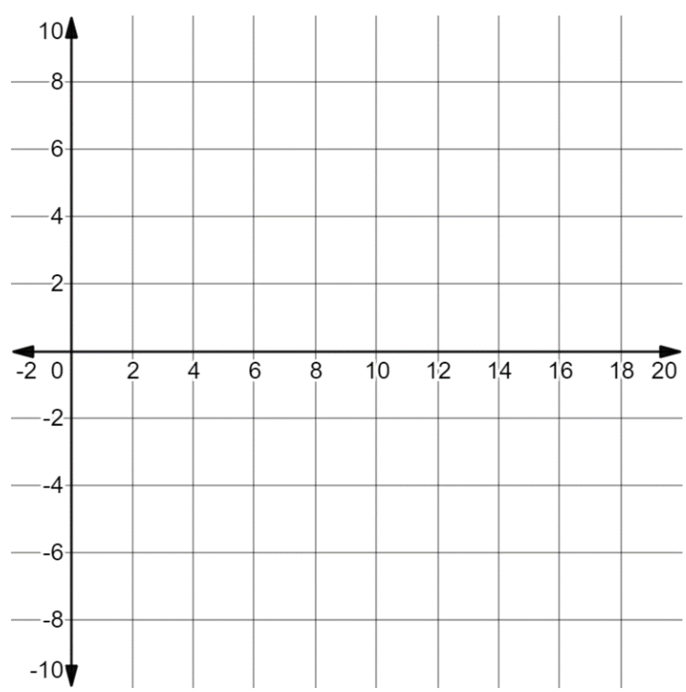
2. The diagram below shows a possible model of Felix's fence with sides labeled as x and y.



3. Recall Felix had 30 feet of fencing to use. Write a two-variable inequality that Felix could use to find possible dimensions for his fence.

4. Explain how to determine which inequality symbol to use.

5. Graph the inequality.



Drew was applying for a personal loan to pay for home improvement projects. His bank requires a minimum of $\$1{,}000$ between his checking and savings accounts to be considered for a loan.

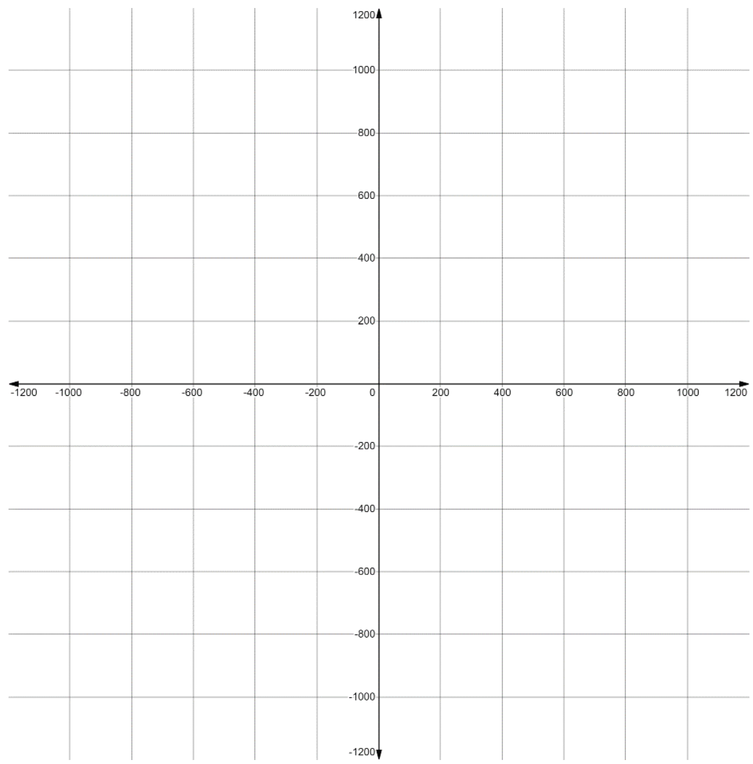
6. Define two variables for this situation.

   Let $x =$ ____

   Ley $y =$ ____

7. Write an inequality in terms of x and y that could be used to determine whether Drew meets the minimum requirement for the loan.

8. Graph the inequality.



9. Complete the statement.

   To be a viable solution in this context, the x and y values must both be rounded to _____ decimal places.

10. Drew overdrew his checking account, which now has a balance of $-\$350.00$. He has $\$1{,}452.65$ in his savings account. Does he meet the minimum requirement?

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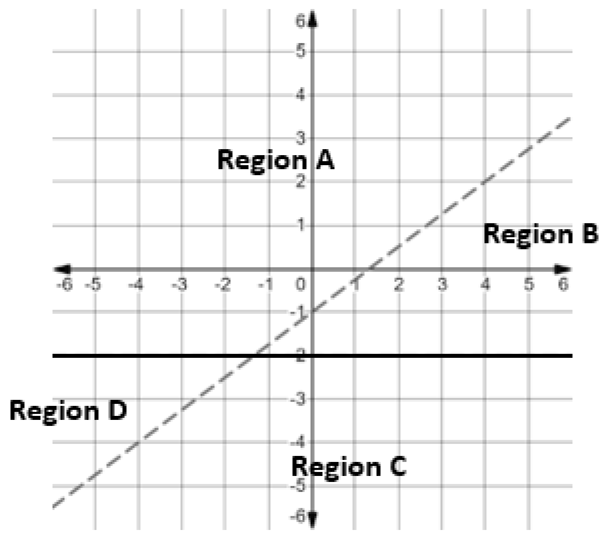
Consider the system of inequalities, $\begin{cases} y \geq -2 \\ 3x - 4y < 4 \end{cases}$.

The boundary lines of both inequalities in the system are shown on the coordinate plane.



1. The boundary lines separate the plane into four regions, labeled A - D. Which region includes the solution set of the system of inequalities?

2. Shade the region that includes the solution set of the system of inequalities.

3. Select a test point in the shaded region of the solution set.

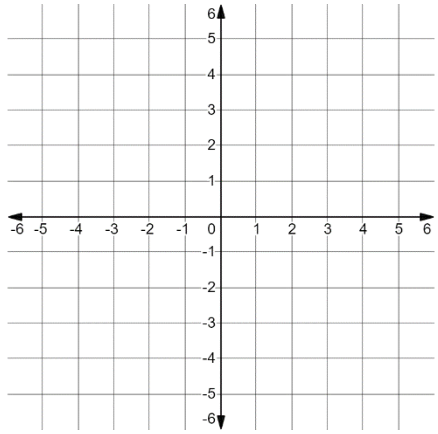
4. Substitute the $x -$ values and the $y -$values of the test point to show that the ordered pair satisfies both inequalities.

$y \geq -2$	$3x - 4y < 4$

5. Your friend says that the point $(3, -2)$ is a solution to the system of inequalities. Explain if you agree or not.

Consider the system of inequalities, $\begin{cases} y + 3 > 0.2(x + 5) \\ y - 4 \le -2(x + 1) \end{cases}$.

6. Graph the boundary lines of the system.

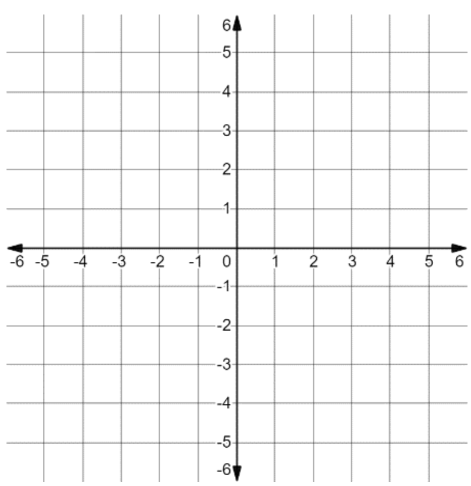


Use substitution to determine whether the given ordered pairs are solutions to the system.

	$\boldsymbol{y + 3 > 0.2(x + 5)}$	$\boldsymbol{y - 4 \le -2(x + 1)}$	**Solution to the system?**
7. $(-3,4)$			○ Yes ○ No
8. $(5,2)$			○ Yes ○ No

9. Complete the graph by shading the region included in the solution set of the system.

10. Graph the solution set of the system of inequalities: $\begin{cases} x \le 2 \\ y > \frac{3}{4}x + 2 \end{cases}$.



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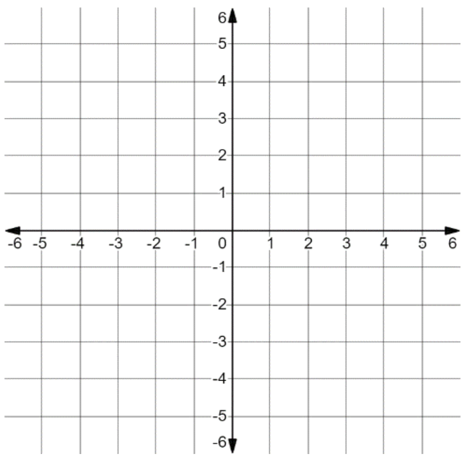
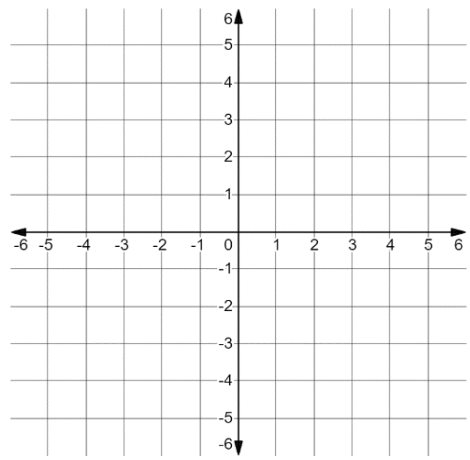
Graph the solution set for each system of inequalities on the coordinate grid.

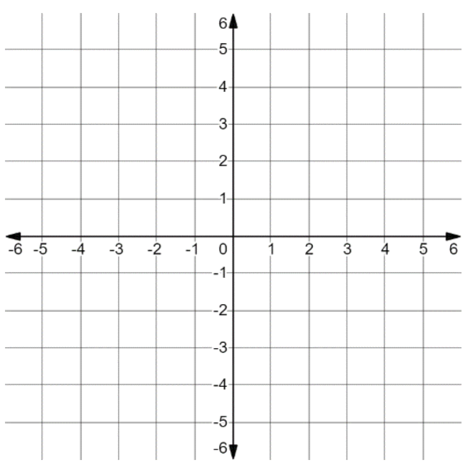
1. $\begin{cases} y < 3 \\ 2x + 3y > 6 \end{cases}$



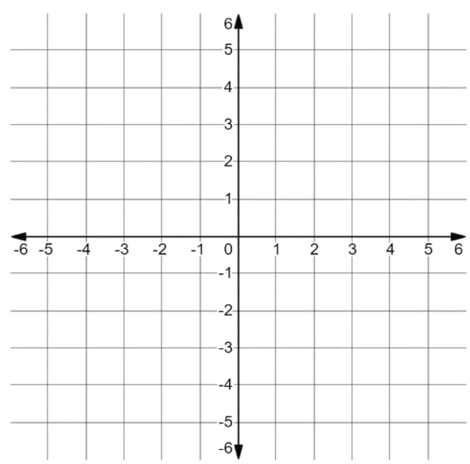
2. $\begin{cases} x \geq -2 \\ 2x - 4y > 8 \end{cases}$



3. $\begin{cases} y \geq 0.8x + 3 \\ y - 2 < 0.8(x - 2) \end{cases}$

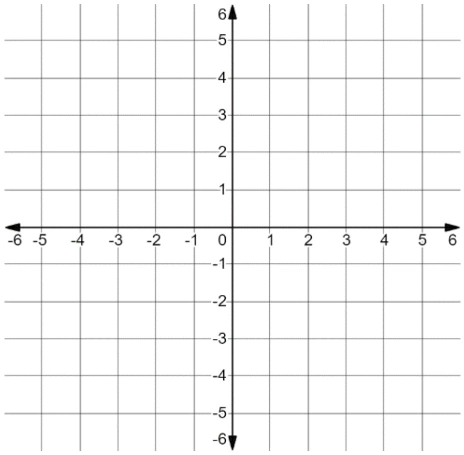


4. $\begin{cases} y < \frac{3}{2}x + 1 \\ y - 1 < -0.5(x - 6) \end{cases}$

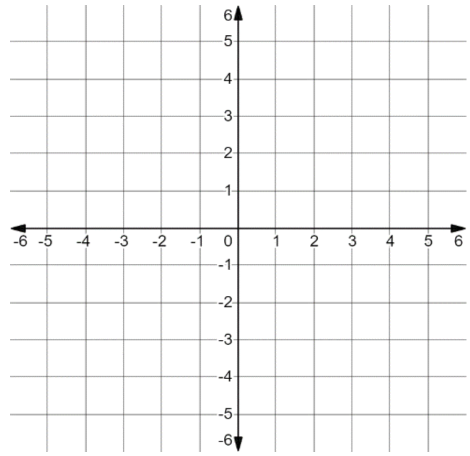


5. From problems 1-4, identify the graph that has no solution and explain your reasoning.

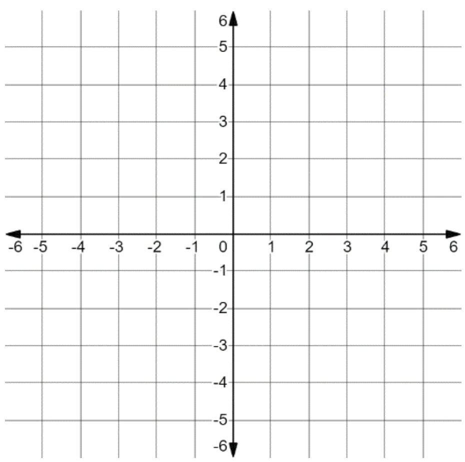
6. $\begin{cases} y > -5 \\ y + 4 \leq \frac{3}{4}(x + 4) \end{cases}$



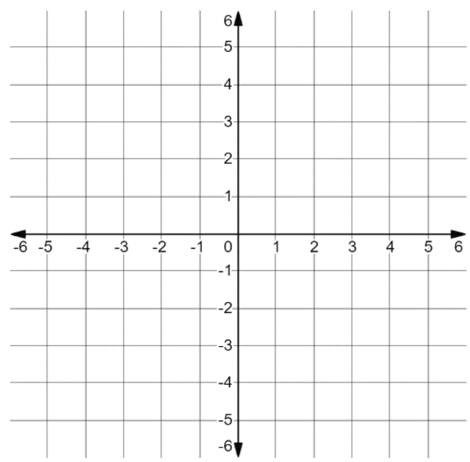
7. $\begin{cases} x \leq 1 \\ y > -\frac{5}{3}x - 3 \end{cases}$



8. $\begin{cases} y \geq -\frac{1}{3}x + 1 \\ y < -3x - 2 \end{cases}$



9. $\begin{cases} x \geq 4 \\ y > -4 \end{cases}$



10. $\begin{cases} -4x - 2y \geq 10 \\ y \geq -\frac{5}{4}x - 3 \end{cases}$